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Learning Objectives:

- Introduction
- Structure
- Distribution
- Metabolism
- Biological functions
- Disorders

VITAMIN D IS REALLY A HORMONE

- A vitamin is defined as an organic compound that is required in the diet in small amounts for the maintenance of normal metabolic integrity.

- However, vitamin D, which is formed in the skin on exposure to sunlight, do not strictly comply with this definition.

- Only when sunlight exposure is inadequate, a dietary source required.

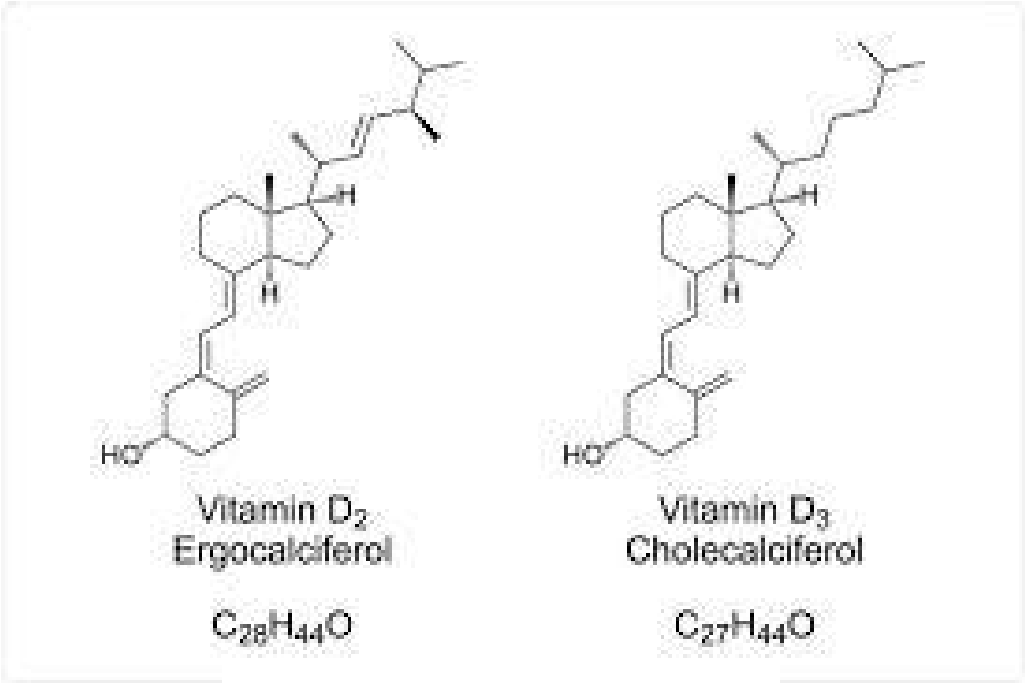
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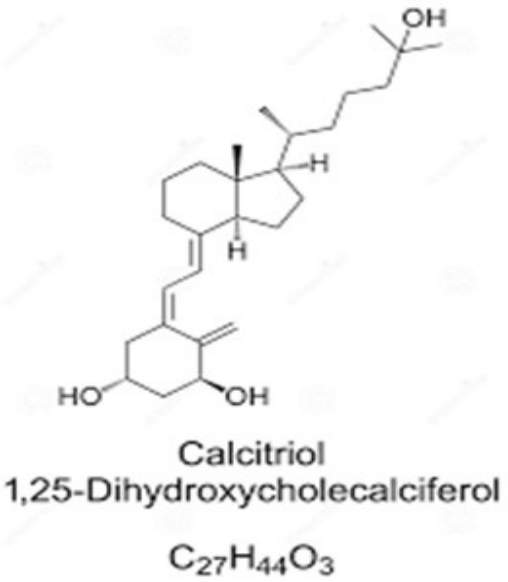
- Its main function is in the **regulation of calcium absorption** and homeostasis.
- It also has a role in regulating cell proliferation and differentiation.
- Higher intake considerably **reduces the risk of insulin resistance, obesity, and the metabolic syndrome**, as well as various **malignant neoplasms**.
- Deficiency, leading to **rickets in children and osteomalacia in adults**.

STRUCTURE

- The D vitamins are a group of sterols.

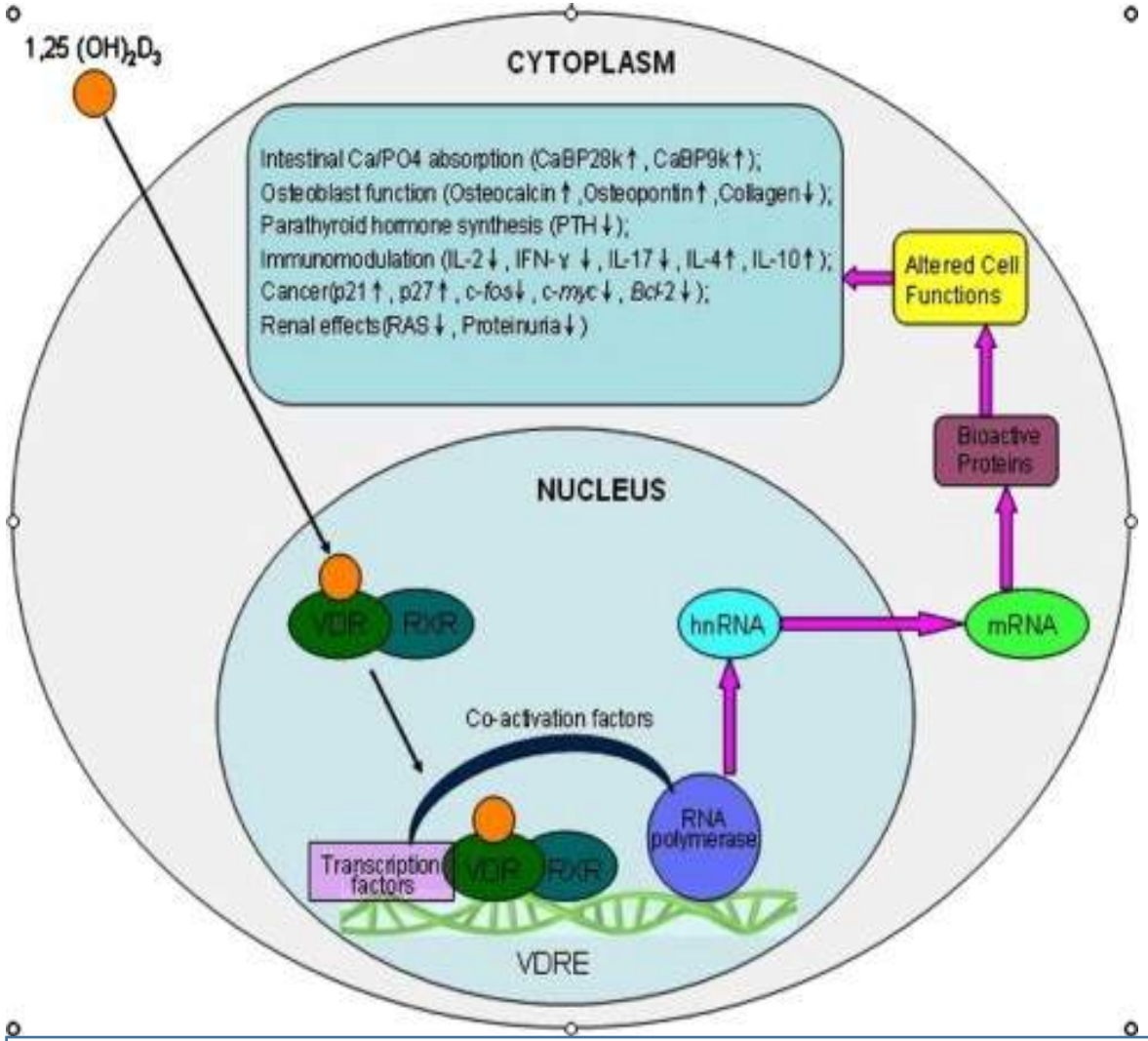
- The active molecule, 1,25-dihydroxycholecalciferol ([1,25-diOH-D3] calcitriol).





VITAMIN D RECEPTOR (VDR)

- It acts like a steroid hormone.
- The active molecule, 1,25-dihydroxycholecalciferol ([1,25-diOH-D3] calcitriol), binds to nuclear receptors.
- The 1,25-diOH-D3–receptor complex interacts with DNA in the nucleus of target cells and enhances gene expression.



VDRE: Vitamin D responsive elements, RXR: Retinoid X receptor

SOURCES OF VITAMIN D

- Safe sun exposure
- Very few foods in nature contain vitamin D:
  - ❑Flesh of fatty fish (salmon and tuna)
  - ❑Fish liver oils
  - ❑Cheese
  - ❑Egg yolks
  - ❑Mushrooms
  - ❑Fortified milk and juices provide most of the vitamin D.



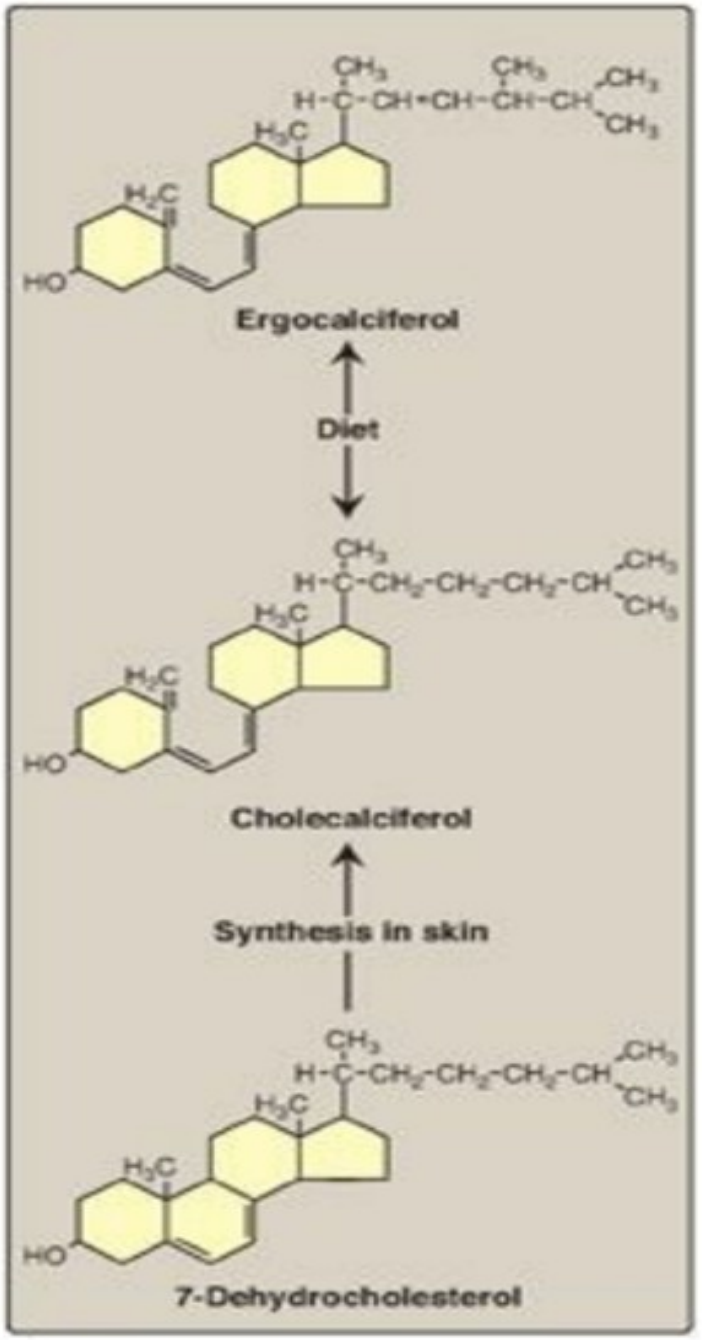
Distribution of vitamin D

1. Endogenous vitamin precursor:

- 7-Dehydrocholesterol, an intermediate in cholesterol synthesis, is converted to cholecalciferol in the dermis and epidermis of humans exposed to sunlight and transported to liver bound to vitamin D–binding protein.

2. Diet:

- Ergocalciferol (vitamin D2), found in plants, and cholecalciferol (vitamin D3), found in animal tissues, are sources of preformed vitamin D activity.
- Dietary vitamin D is packaged into chylomicrons.
- [Note: Preformed vitamin D is a dietary requirement only in individuals with limited exposure to sunlight.]



Metabolism of vitamin D

- **Formation of 1,25-dihydroxycholecalciferol:**

- Vitamins D2 and D3 are converted to active form by two sequential hydroxylation reactions.

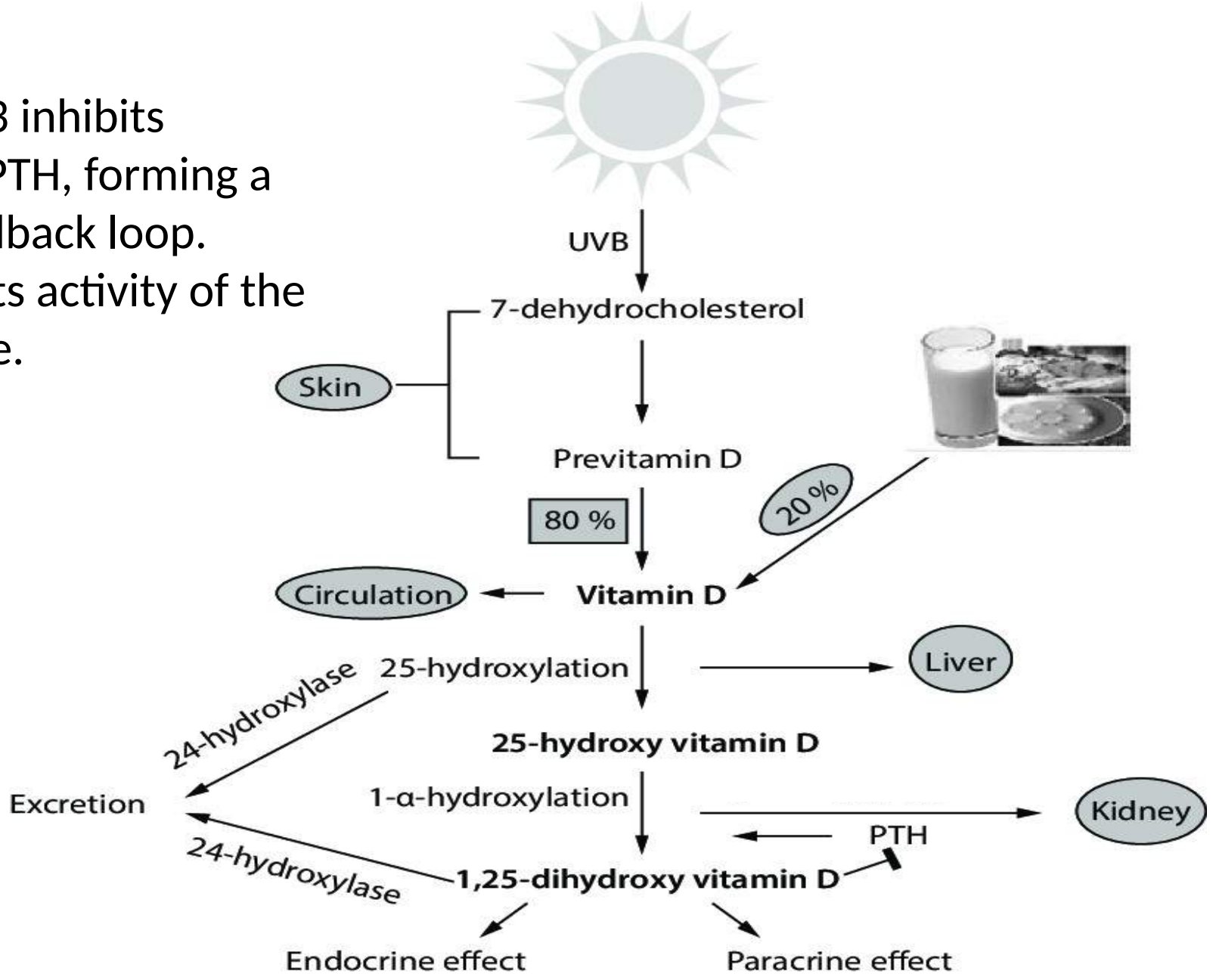
- The first hydroxylation occurs at the **25 position and is catalyzed by a specific 25-hydroxylase in the liver.** This is released into the circulation bound to a vitamin D–binding globulin. The product of the reaction, 25-hydroxycholecalciferol ([25-OH-D3], calcidiol), is the **predominant form of vitamin D in the plasma** and the major storage form of the vitamin.

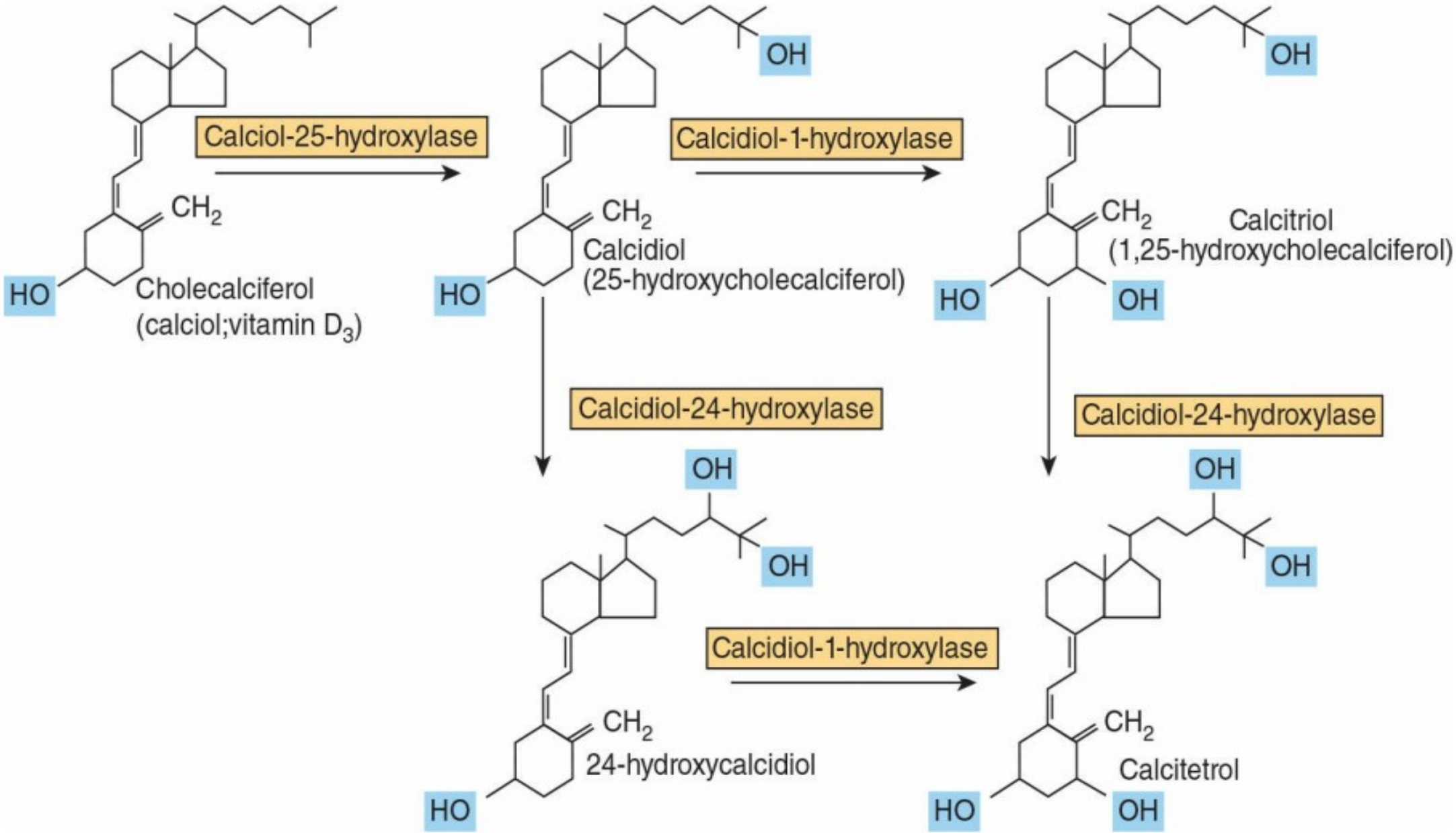
- 25-OH-D3 is further hydroxylated at the **1 position by 25- hydroxycholecalciferol 1-hydroxylase found primarily in the kidney**, resulting in the formation of 1,25-diOH-D3 (calcitriol)
- Or it may undergo in 24-hydroxylation to yield a probably inactive metabolite, 24,25-dihydroxyvitamin D (24-hydroxycalcidiol)..

Regulation:

1,25-diOH-D3 inhibits synthesis of PTH, forming a negative feedback loop.

It also inhibits activity of the 1-hydroxylase.



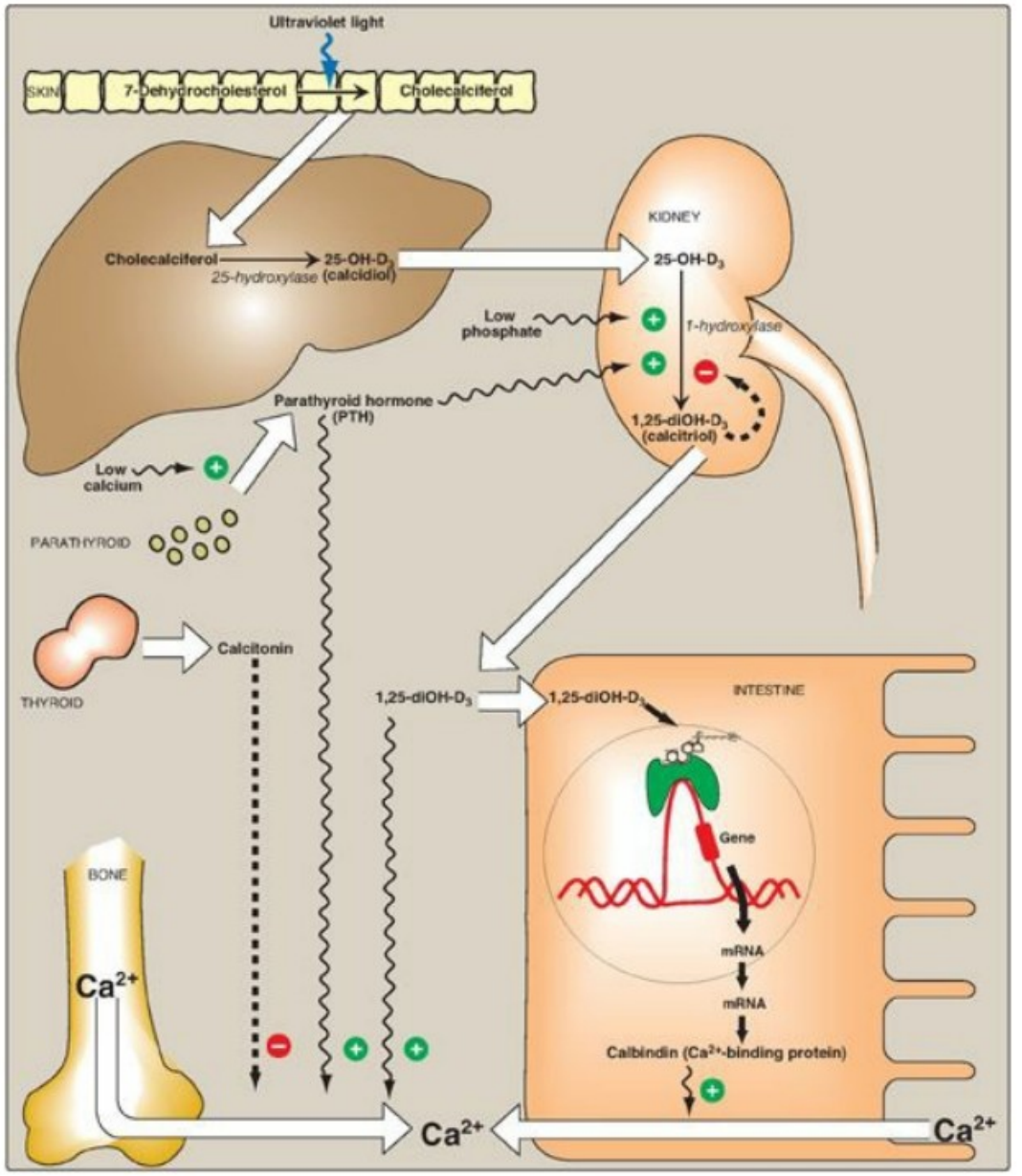

Calciol-25-hydroxylase
Calcidiol-1-hydroxylase
Calcidiol-24-hydroxylase
Calcidiol-24-hydroxylase
Calcidiol-1-hydroxylase
HO
CH2
Cholecalciferol
(calciol;vitamin D3)
OH
Calcidiol
(25-hydroxycholecalciferol)
Calcitriol
(1,25-hydroxycholecalciferol)
24-hydroxycalcidiol
Calcitetrol

Regulation of 25-hydroxycholecalciferol 1-hydroxylase:

- 1,25-diOH-D3's formation is tightly regulated by the level of plasma phosphate and calcium ions.

- Low plasma calcium levels trigger the secretion of parathyroid hormone (PTH) from parathyroid gland.
- PTH upregulates the 1-hydroxylase. Thus, hypocalcemia caused by insufficient dietary calcium results in elevated levels of plasma 1,25-diOH-D3.

- Low plasma phosphate directly stimulates 1-hydroxylase activity .



Recommended Dietary Allowances (RDAs) for Vitamin D

Age	Male	Female
0–12 months*	400 IU (10 mcg)	400 IU (10 mcg)
1–13 years	600 IU (15 mcg)	600 IU (15 mcg)
14–18 years	600 IU (15 mcg)	600 IU (15 mcg)
19–50 years	600 IU (15 mcg)	600 IU (15 mcg)
51–70 years	600 IU (15 mcg)	600 IU (15 mcg)
>70 years	800 IU (20 mcg)	800 IU (20 mcg)

Institute of Medicine, and Endocrine Society Recommended Vitamin D intake

UL: The Tolerable Upper Intake Level is the maximum daily intake unlikely to cause harmful effects on health. The UL for vitamin D for adults and children is **4,000 IU**.

Function of vitamin D

- The principal function of vitamin D is to **maintain adequate plasma calcium concentration.**

- Calcitriol achieves this in three ways:;
  - 1) it increases intestinal absorption of calcium
  - 2) it reduces excretion of calcium (by stimulating reabsorption in the distal renal tubules); and
  - 3) stimulating resorption (demineralization) of bone when blood calcium is low.

OTHER ACTIONS OF VITAMIN D

- In addition, calcitriol is involved in wide range of biological actions:
  - Inhibiting cellular proliferation and inducing terminal differentiation
  - Inhibiting angiogenesis
  - Stimulating insulin production and secretion
  - Inhibiting renin production
  - Remove dead or dying cells
  - Higher levels of serum 25(OH)D leads to lower incidence of cancers, including prostate and colorectal cancer.
  - Synthesis and secretion of parathyroid and thyroid hormones

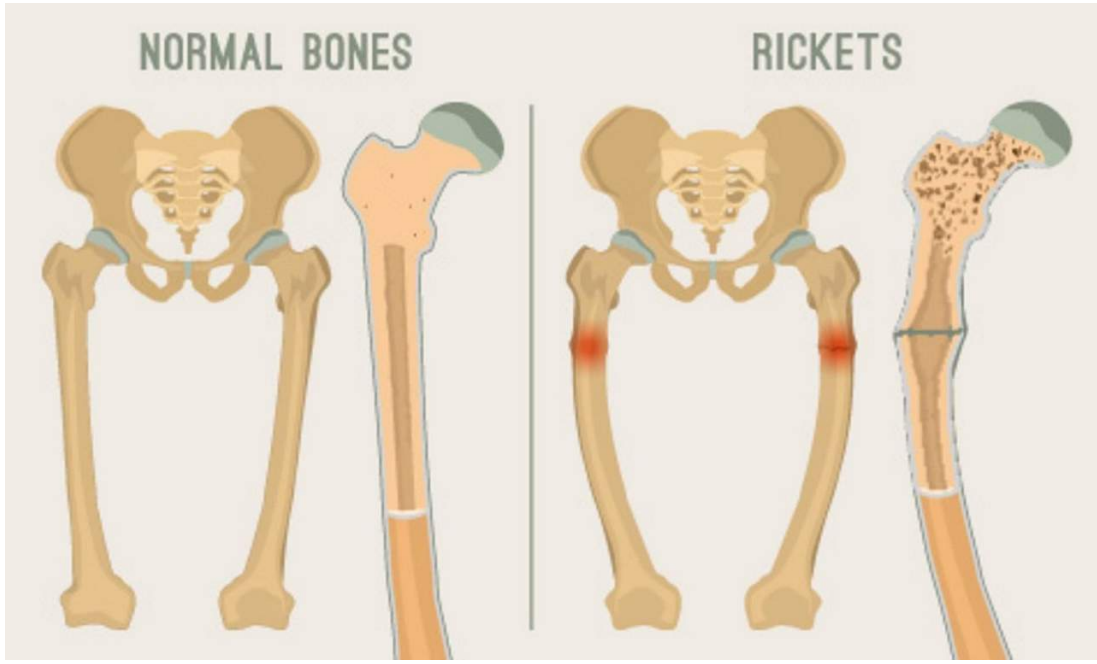
Vitamin D Deficiency Affects Children & Adults

- Vitamin D deficiency causes a net demineralization of bone, resulting in rickets in children and osteomalacia in adults.

- In Rickets the bones of children are undermineralized as a result of poor absorption of calcium, and is characterized by the continued formation of the collagen matrix of bone, but incomplete mineralization results in soft, pliable bones.

- Osteomalacia in adults results from the demineralization of bone, especially in women who have little exposure to sunlight, and especially after several pregnancies. The demineralization of pre-existing bones increases their susceptibility to fracture.




NORMAL BONES
RICKETS



Renal osteodystrophy:

- Chronic kidney disease causes decreased ability to form active vitamin D as well as increased retention of phosphate, resulting in hyperphosphatemia and hypocalcemia.

- The low blood calcium causes a rise in PTH and associated bone demineralization with release of calcium and phosphate.

Vitamin D Is Toxic in Excess

- Some infants are sensitive to intake of vitamin D as low as 50 µg/d, resulting in an elevated plasma concentration of calcium. This can lead to contraction of blood vessels, high blood pressure, and calcinosis—the calcification of soft tissues. It may be due to genetic defects of calcidiol 24-hydroxylase, the enzyme that leads to inactivation of the vitamin.

- Although excess dietary vitamin D is toxic, excessive exposure to sunlight does not lead to vitamin D poisoning b/c excess vitamin D produced in the skin is converted to inactive forms.

- High doses (100,000 IUs for weeks or months) can cause loss of appetite, nausea, thirst, and stupor.

THANK YOU